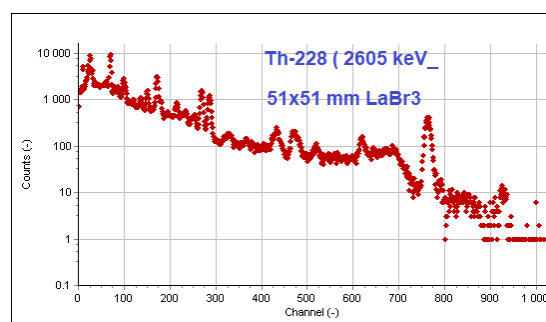
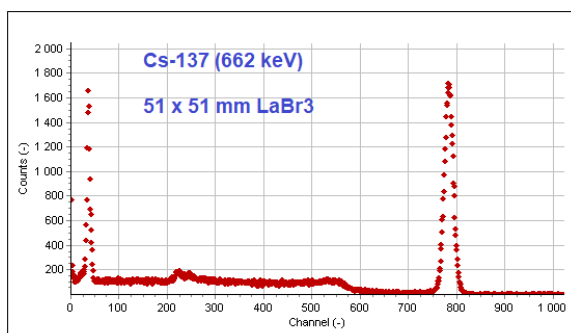


High resolution Ce doped LaBr₃ scintillators

LaBr₃:Ce scintillation crystals offer an energy resolution at 662 keV superior to all other available scintillators. The material is fast and suitable for high count rates. It suffers from the background of La-138 via beta emission and some gamma lines (absent in CeBr₃). In addition some alpha lines are present above 2.5 MeV. These are associated with the decay chain of Ac-227.

Density	:	5.07 g / cc
Typical Ce concentration	:	0.5 mole percent
Maximum emission	:	370 nm
Decay time (typical)	:	16-20 ns (size dependent)
Refractive index	:	2.09 (380 nm)
Photons yield	:	Approx. 66.000 / MeV
Hygroscopic	:	Yes
Typical energy resolution @ 662 keV :		3.0 % FWHM



Energy (keV)	Typical resolution LaBr ₃	Typical resolution NaI(Tl)
30 (129-I)	15 %	18 %
59.5 (241-Am)	10 %	11 %
122 (57-Co)	6.5 %	8.5 %
662 (137-Cs)	3 %	7 %
1332 (60-Co)	2.5 %	5.5 %
2600 keV (Th-228)	2 %	4.0 %

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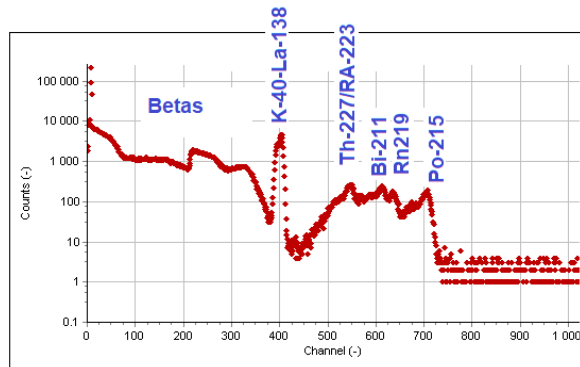
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Maximum dimensions : 76 mm diameter, 76 mm high



Typical LaBr₃ background spectrum (unshielded 51 x 51 mm)

The background in the beta/gamma spectrum is approx 2.5 c/s/cc

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